



Instituto de Tecnologia e Liderança

Public Report

Intelligent Platform for Social Engagement

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1. Document Version Control

Date	Version	Changes	Author
04/27/2025	V1	Initial template, Introduction and week 1 logbook	Ana Clara Zaidan
05/12/2025	V2	Weekly logbook - week 2	Ana Clara Zaidan
05/13/2025	V03	Weekly logbook - week 3	Ana Clara Zaidan
05/15/2025	V04	Weekly logbook - week 4	Ana Clara Zaidan
06/20/2025	V05	Weekly logbook - weeks 5, 6, 7, 8, 9, 10	Ana Clara Zaidan
06/24/2025	V06	Artifacts summary	Ana Clara Zaidan

2. Introduction

This document consolidates the progress and artifacts developed for the Project titled "Intelligent Platform for Social Engagement". The project partner is the social organization "Olhar de Bia", whose mission is to end poverty through education. Ms. Beatriz Martins, the NGO's founder, acts as the Product Owner and focal point.

The main problem addressed is the difficulty in effectively connecting individuals willing to volunteer with social organizations needing collaboration. Many people give up volunteering because they don't know how to find projects or are unaware of existing institutions. Conversely, Civil Society Organizations (CSOs) face challenges in publicizing opportunities and selecting suitable volunteers, alongside difficulties in managing volunteer work. Research conducted in São Paulo (CHANG, 2022) indicated that 48.6% of respondents had never participated in volunteer work, citing reasons such as difficulty finding projects and lack of knowledge about institutions.

Given this scenario, the main objective of the project is to develop a web platform that centralizes volunteer opportunities and donation campaigns, using Artificial Intelligence to connect volunteers with causes and events best aligned with their profiles and interests.

For partner organizations, like "Olhar de Bia," the platform aims to:

- Increase the reach of potential volunteers and donors.
- Improve the "fit" between volunteers and the organization's needs/causes.

From a social impact perspective, the main objectives are:

- Stimulate participation in diverse social causes.
- Indirectly contribute to guiding resources - such as time and donations - toward organizations, thereby generating social impact.

The project started on February 4th, 2025, and is scheduled for completion on December 19th, 2025. The project is divided into four "modules" of 10 weeks each, and this report outlines the progress made in the second module, including the Technology Stack, the Solution Architecture with C4 models, Data Modeling and Environment and Repository Setup.

3. Tech Stack

The solution is designed with a **three-main-layer structure** including a frontend, a backend API, and a lightweight recommendation service. The selection of technologies was based on their robustness, community support, compatibility with cloud deployment (AWS), and familiarity with the tools.

The **Frontend** application is responsible for user interaction and is developed using:

- **TypeScript**: A statically-typed language that enhances JavaScript.
- **Angular**: A web application framework designed for building dynamic and scalable single-page applications (SPAs).
- **Sass**: A CSS preprocessor for writing maintainable and reusable styles.

The **Backend** API manages business logic and data persistence, utilizing:

- **ASP.NET Core:** Chosen for its performance and scalability in enterprise backends, used for the Web API, business logic, and authentication.
- **C#:** The primary programming language.
- **PostgreSQL:** Serves as the relational database management system for persistent data storage.

The **AI-Based Recommendation Service** is built to provide personalized event recommendations, leveraging Python's data science ecosystem. Its key technologies include:

- **Python:** The primary programming language for the service.
- **Flask:** A lightweight web framework used to expose the recommendation API.
- **Pandas:** A data analysis and manipulation library.
- **scikit-learn:** Provides machine learning algorithms for the recommendation logic.
- **spaCy:** Used for natural language processing to interpret user interests.
- **pytest:** Employed for unit testing Python-based modules.

To ensure maintainability and reliability, an **Observability Stack** has been adopted. This includes:

- **OpenTelemetry:** A unified framework for collecting traces, logs, and metrics across services, simplifying instrumentation and improving end-to-end visibility.
- **Prometheus:** Responsible for collecting real-time system and service metrics, such as API response times and error rates.
- **Grafana:** Serves as the real-time dashboard for visualizing metrics collected by Prometheus, enabling monitoring of system health and performance.
- **ELK Stack** (Elasticsearch, Logstash, Kibana): Provides centralized logging, indexing, and querying, facilitating error tracing and debugging.

For **DevOps & Infrastructure**, the project incorporates:

- **Docker:** Adopted as the containerization solution to ensure environment consistency, portability, faster deployments, easier scaling, and better reproducibility of environments.
- **GitHub Actions:** Used for CI/CD automation.
- **AWS (Amazon Web Services):** The chosen cloud platform for hosting the entire project infrastructure.

4. Solution Architecture

The **C4 model**, developed by Simon Brown, is utilized to describe the solution architecture across multiple levels of abstraction, facilitating clear communication among stakeholders (BROWN, s.d.). This report focuses on the first two levels: Context and Containers.

Level 1: Context Diagram

The Context Diagram provides a **high-level overview** of how the platform interacts with its primary users and external systems.

- **Volunteers** (Person) look for and subscribe to social events using the platform.
- **Non-Governmental Organizations (NGOs)** (Person) publish social events on the platform.
- The **Intelligent Platform for Social Engagement** (Software System) enables NGOs to share events and volunteers to enroll.
- An **AI Recommendation Service** (External API) is used by the platform to suggest events based on user interests.
- An **Observability & Monitoring Service** (External Service) collects metrics and logs from the platform.

Level 2: Container Diagram

The Container Diagram offers a **deeper view of the platform's internal structure**, detailing the main containers (applications, services, and databases) and their key technologies.

- The **Single-Page Application** (Container: TypeScript and Angular) serves as the frontend interface, providing all functionality to customers via their web browser.
- The **API Application** (Container: .NET using C#) provides functionality via a JSON/HTTPS API, manages business logic, and handles data processing.
- A **Database** (Container: PostgreSQL) stores platform data, including user profiles and event details.
- The **AI Recommendation Service** (Container: Flask, Python) generates personalized event recommendations and receives data from the database.
- The **Observability & Monitoring Service** (Container) is composed of:
 - **Prometheus** for metrics collection.
 - **Grafana** for data visualization of metrics.
 - **OpenTelemetry** for distributed tracing across containers.
 - **ELK** (Elasticsearch, Logstash, Kibana) for centralized log management. The API Application sends metrics to Prometheus and traces to OpenTelemetry, and logs to ELK.

5. Data Modeling

Effective data modeling is essential for designing robust, scalable, and maintainable information systems. According to the American National Standards Institute (ANSI), the data modeling process is typically divided into three levels of abstraction: conceptual, logical, and physical data models (WIKIPEDIA, 2024). For this stage of development, it was decided to focus on the development of a **Conceptual Data Model**, followed by a **Logical Data Model**.

The **Conceptual Model** captures the scope and semantics of the data, identifying core entities and their relationships. It serves as a foundation for understanding data requirements and aligning them with business objectives. For the platform, the conceptual model identifies two main user roles: volunteers and company representatives (NGOs). Key entities identified in the conceptual model include:

- **User:** Represents volunteers and company representatives, with attributes like name, email, birthdate, address, and associated companies.
- **Company:** Represents organizations that can create events, with information such as biography, contact details, and social media URLs.
- **Event:** Created by a single company and open to multiple volunteer participants.
- **Keyword:** Tags that describe user interests and event topics.

The conceptual model is visually represented by an Entity-Relationship Diagram (ERD), illustrating that companies can publish multiple events, each event belongs to one company, and users can subscribe to events and define preferences via keywords.

Next, the **Logical Data Model** was derived from the conceptual model. This model refines and structures the information into a relational schema, detailing tables, attributes (columns), primary keys, foreign keys, and relationships, while remaining independent of physical storage concerns. This detailed organization ensures efficient data management and forms the basis for the physical database implementation.

The theoretical foundations of relational databases, including ACID properties (Atomicity, Consistency, Isolation, Durability), data integrity and consistency through primary and foreign keys, scalability, operational flexibility, and security with detailed access controls, justify the choice of this paradigm. Additionally, the normalization

process ensures optimized data structures by reducing redundancy and increasing data integrity and performance.

6. Environment and Repository Setup

The project's environment setup includes the utilization of **Dev Containers**. Dev Containers are a standardized and configurable way of creating consistent and isolated development environments. They allow project dependencies, tools and configurations to be packaged in a container, ensuring that all developers have the same environment, regardless of the operating system or tools installed on their local machines.

For this project, we will be using GitHub Codespaces, which is a cloud-based platform that provides instant and configurable development environments directly in the browser or via VS Code. GitHub Codespaces uses Dev Containers to provision a ready-to-use development environment, with all the tools and dependencies pre-configured, eliminating the need to manually configure the local environment.

7. Weekly Logbook (Weekly Notes)

Designed to systematically record the project's weekly progress, the Weekly Logbook highlights the activities performed, documents challenges encountered, details solutions applied, and reflects on critical learnings. Additionally, it serves to outline planned next steps, ensuring continuous improvement and alignment with project goals.

Week 1 (04/22/2025 to 04/27/25):

Activities Performed:

- Held the initial meeting with the advisor on April 25th.
- Defined my main focus for this module and identified the areas in which I am most interested in developing skills.
- Received instructions regarding the artifacts to be delivered for this module, as well as feedback for improvements based on the previous module.
- Reviewed the project plan submitted in the previous module and applied the necessary updates regarding the project scope, features, and artifacts to be delivered.

Challenges Encountered:

- Aligning project ideas with the course requirements and structure.
- Adapting the current module to address pending tasks from the previous module.

Solutions Applied:

- Engaged in open discussions with the advisor regarding project interests and course expectations.
- Began utilizing Trello more actively.
- Improved personal organization by creating more “to-do” lists rather than only maintaining “pending tasks” lists.

Learnings:

- Gained a clearer understanding of the expectations for this module concerning the required artifacts.
- Experienced firsthand the importance of maintaining a well-defined, complete, and continuously updated backlog, even when managing a team of one.

Next Steps:

- Keep Trello consistently updated and continue to enhance its use.
- Gradually address outstanding tasks from the previous module, particularly those related to the tech stack and prototyping.

Week 2 (04/28/2025 to 05/04/2025):

Activities Performed:

- Held a meeting with the advisor on April 29th
- Incorporated the pending items from the previous module into my plan.
- Finished updating and reviewing the proposed changes to the roadmap.

Challenges encountered:

- Thinking I won't be able to deliver what was proposed on time.
- Insecurity regarding the scope of the project and technical capacity.

Solutions Applied:

- Possibility of focusing on concept tests instead of a full implementation, with intelligent prioritization.

Learnings:

- The project is not immutable and we can adapt it to our needs as we go along.
- It is necessary to keep priority items and main focuses in mind in order to channel more energy and time towards them.

Next steps:

- Better understand my focus in developing the project.
- Start delivering the new tasks that have been included in the roadmap and ensure the continuity of the project's development, avoiding the creation of new pendency backlog tasks.

Week 3 (05/05/2025 to 05/11/2025):

Activities Performed:

- Held a meeting with the advisor on May 9th.
- Structured the project's tech stack.
- Documented the rationale behind the selection of each technology.
- Designed diagrams categorizing and detailing the stack components.
- Made minor adjustments to the project plan, based on feedback received from the Sprint 1 deliverable.

Challenges encountered:

- Balancing the ambition of designing a robust architecture with the realistic limitations of project scope and available resources.
- Uncertainty regarding available cloud resources and infra.

Solutions applied:

- Opted to propose a robust solution, relying on technologies that are familiar or of strong interest for further learning.
- Focused on selecting technologies that I have interest in learning more.

Learnings:

- Discovered StackShare, a tool that facilitates the visualization and comparison of tech stacks.

Next steps:

- Validate the proposed stack with the advisor.
- Study about C4 diagrams.
- Begin designing the solution architecture based on the defined stack using C4.

Week 4 (05/12/2025 to 05/18/2025):

Activities Performed:

- Held a meeting with the advisor on May 13th.
- Studied about the C4 Model, including its levels of abstraction and detail.
- Developed the Context and Container diagrams as part of the solution architecture.
- Learned to use *Mermaid* as a tool to generate architecture diagrams.

Challenges Encountered:

- Limited prior experience with the C4 Model and its practical application.
- Difficulty in retrieving and consulting scientific literature and academic articles.

Solutions Applied:

- Focused on consulting the official documentation and reference material from Simon Brown's *"The C4 Model for Visualising Software Architecture."*

Learnings:

- Gained proficiency in generating diagrams using *Mermaid* and refining them for more accuracy.
- Acquired understanding of the C4 Model.

Next Steps:

- Validate the proposed architecture and tech stack to finalize the technological direction of the project.
- Prepare for the upcoming implementation phase by consolidating architectural decisions.

Week 5 (05/19/2025 to 05/25/2025):**Activities Performed:**

- Held a meeting with the advisor on May 20th.
- Reviewed and refined the prototype originally proposed in Module 1.
- Studied about relational (SQL) and non relational (NoSQL) databases.

Challenges Encountered:

- Faced difficulties in managing time among project tasks and personal commitments.

Solutions Applied:

- Focused efforts on strengthening the foundational knowledge of relational databases.

Learnings:

- Gained a clearer understanding of the advantages and limitations of relational databases.
- Developed awareness of appropriate scenarios for adopting NoSQL databases, including scalability and flexibility considerations.

Next Steps:

- Develop the logical and conceptual data models.
- Document the rationale behind database modeling choices.

Week 6 (05/26/2025 to 06/01/2025):

Activities Performed:

- Held a meeting with the advisor on May 27th.
- Developed the conceptual and logical data models for the system.

Challenges Encountered:

- Encountered persistent challenges in time management.
- Faced knowledge gaps related to Entity-Relationship Diagrams (ERD) and Entity-Relationship Modeling (ERM).

Solutions Applied:

- Adopted a strategy of task prioritization to focus on critical deliverables.
- Derived the logical data model directly from the conceptual model, improving the modeling process.

Learnings:

- Gained clarity on the distinct purposes of conceptual, logical, and physical data models.
- Understood how to optimize the modeling workflow by initiating with the conceptual model.

Next Steps:

- Deepen documentation regarding the choice of databases and modeling decisions.
- Develop the physical data model.
- Refine the bibliography and academic references supporting the project.

Week 7 (06/02/2025 to 06/08/2025):

Activities Performed:

- Held a meeting with the advisor on June 3rd.
- Addressed pending items related to the data modeling tasks.
- Mapped all required activities for setting up the development environment and repository structure to support future implementation.

Challenges Encountered:

- Designing an environment setup that addresses the project's needs being realistic given the resources.

Solutions Applied:

- Decided to consolidate the entire project into a single repository to simplify version control and project documentation.
- Adopted Dev Containers for configuring the environment.

Learnings:

- Acquired knowledge about Dev Containers and their benefits in maintaining development environments.

Next Steps:

- Implement the Dev Container setup using *GitHub Codespaces*.
- Install all necessary dependencies and technologies required for the environment setup.

Week 8 (06/09/2025 to 06/15/2025):

Activities Performed:

- Held a meeting with the advisor on June 10th.

- Deepened knowledge about *Dev Containers*, including core concepts and its advantages.
- Studied the file structure and configuration of *GitHub Codespaces*.
- Defined the overall folder and repository structure, organizing the project into key modules: Frontend, Backend, Machine Learning, Observability, and Infrastructure.
- Studied about containerization principles and Docker.
- Installed and configured the main technologies required for the project, including Angular, Node.js, Go, and Python.

Challenges Encountered:

- Faced difficulties in configuring *Docker-in-Docker* within the Dev Container environment.
- Encountered issues accessing the Dev Container via Visual Studio Code.

Learnings:

- Understood how containerized environments simplify the initial setup and maintenance of development workspaces

Next Steps:

- Document the decisions and rationale behind the chosen environment configuration.
- Setting up the Observability module environment.

Week 9 (06/16/2025 to 06/22/2025):**Activities Performed:**

- Held a meeting with the advisor on June 17th.
- Developed the final presentation for the module, focused on the project partner.
- Initiated the structuring of this public project report.
- Refined the contents of this Weekly Logbook to ensure clarity and accuracy.

Challenges Encountered:

- Deciding which topics and deliverables to prioritize in the final presentation.

Solutions Applied:

- Oriented the presentation to be centered on the project partner, focusing on delivering insights and value relevant to the partner's expectations and needs.

Learnings:

- Improved time management and decision making regarding deliverable prioritization.

Next Steps:

- Finalize the public report for the current module.
- Continue advancing the environment setup to ensure readiness for the implementation phase.

Week 10 (06/23/2025 to 06/29/2025):**Activities Performed:**

- Held a meeting with the advisor on June 24th.
- Completed the environment setup for the Observability module.
- Documented the technical details of the environment configuration.
- Advanced the development of the public report sections for the current module.

Challenges Encountered:

- Faced difficulties balancing time between completing the public report and finalizing technical artifacts for the sprint.
- Encountered challenges in task prioritization while managing multiple demands.

Solutions Applied:

- Utilized Trello to enhance task management and prioritization.
- Focused efforts on completing deliverables from the current sprint, while deferring tasks or revisions from previous sprints to a later stage.

Learnings:

- Further refined task prioritization strategies, focusing on present sprint objectives over backlog accumulation.

Next Steps:

- Initiate the implementation phase of the project, leveraging all artifacts and documentation developed during the first two modules.
- Conduct iterative processes of review, improvement, and refinement of current deliverables to ensure quality.

8. Weekly Classes Attended

This section lists the mandatory classes attended every Tuesday during the analyzed period.

- **April 22, 2025:** CAREER TRACK – Hacking Cultures and Power Dynamics
- **April 29, 2025:** CAREER TRACK – Hacking Cultures and Power Dynamics: Challenges with Your Peers
- **May 6, 2025:** ENTREPRENEURSHIP TRACK – Product Design: Developing a Product Thesis Based on Market Data
- **May 13, 2025:** CORPORATE TRACK – Innovation, Applied Research, and Digital Transformation: Tools and Emerging Technologies (AI and Blockchain).
- **May 20, 2025:** CORPORATE TRACK – Job Market in Computing and Certifications: Relevant Certifications – Importance, Preparation, and Career Impact
- **May 27, 2025:** CORPORATE TRACK – Intellectual Property and Patents: Registration Processes – How to Register Patents, Trademarks, and Copyrights
- **June 3, 2025:** CAREER TRACK – Hacking Cultures and Power Dynamics: Challenges with Your Boss
- **June 10, 2025:** CAREER TRACK – Hacking Cultures and Power Dynamics: Challenges Within Your Organization.
- **June 17, 2025:** ENTREPRENEURSHIP TRACK – Product Design: Techniques to Validate the Product Thesis Through MVP (Minimum Viable Product).

- **June 24, 2025:** CORPORATE TRACK – Corporate Venture Capital and M&A:
Due Diligence Process

9. References

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